Dear Phillips Academy,

Thank you for allowing me to submit this additional document to my application.

<u>A brief overview of All the nine novel Algorithms I have built to achieve a High compression ratio</u>

1. Sparse event encoding

   The first core idea is to avoid storing every frame in full when most of the visual scene has not changed. Instead of treating video as a continuous stream of complete images, my system tries to store only meaningful changes between frames. This is the base of the compression logic and if a region of the image does not change in a meaningful way, the system should not spend storage and compute resources on it.

2. Proximity correlation and pixel clustering

   The second idea is that changes in video often appear in collective and local groups rather than as isolated random pixels. Its always the truth unless its a very detailed pattern like a galaxy or a star populated sky. For example, when a person speaks, many small pixel changes occur around the mouth rather than across the whole frame. I Instead of recording each changed pixel independently, processor groups nearby changes into clusters and stores compact binary values of those regions. This reduces data size while preserving the location and meaning of the motion. I am also willing to implement a custom binary encryption, so we can drastically reduce the code syntax storage too!

3. Habituation for repetitive motion

   Another major innovation is an algorithm inspired by biological habituation and the ability to stop spending attention on motion that is repetitive and predictable. In the project documents, I use examples such as a rotating fan or a mounted rotating camera that moves continuously in the same pattern. And this was very well documented in the multimedia link I previously attached. If a region repeats the same cycle again and again, the system tries to detect that periodic structure, stop encoding every repeated change, and treat it as habituated motion, and then repeat it with a pixel loop of few bytes in most cases.

4. Looped replacement of repetitive patterns

   Habituation alone would create a false result if a moving fan were simply ignored, because the fan would appear frozen. To solve that, the system attempts to save one minimal repeating cycle of the motion and then replay that loop during reconstruction. In the documents, this is described through a loop mode. And detect the repeating cluster, store one cycle, and reuse it during playback. This is to preserve the visual correctness while reducing the amount of stored data.

5. Motion classification in repetitive versions compared to chaotic versions against information transmission

    A closely related innovation is the attempt to seperate between different kinds of motion rather than treating all motion equally. Repetitive mechanical motion may be compressible as a loop. Chaotic motion, such as fire, water, or turbulence, may look repetitive at first glance but is actually not periodic and may contain meaningful detail. Sparse but important motion, such as a person's mouth while speaking or a small distant object moving in an otherwise still scientific video, must not be discarded. This is mainly achieved through color presence per each frame and comparing them against the area of the relevant pixel density causing the effect.

6. Saliency maps

    The architecture also uses a saliency map in the analysis stage or in the user manual, stage 4. In simple terms, this is a running estimate of which regions deserve storage and compute budget. Regions with little change are low priority. Regions with meaningful, irregular change are high priority. Repetitive high frequency motion may be redirected into loop-based storage instead of full event capture. This is one of the mechanisms intended to help the system decide what matters most in a frame sequence.

7. Mainframe prioritization Algorithm (MPA)

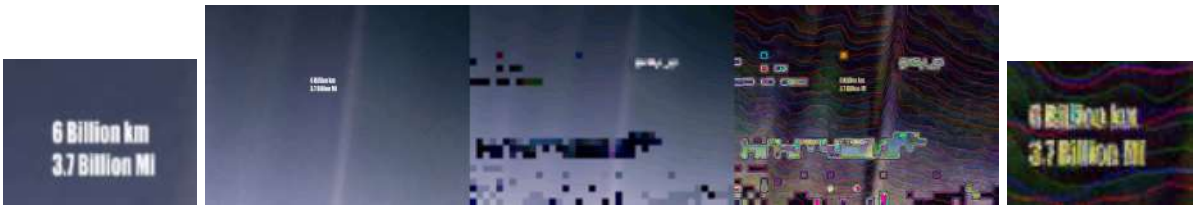
    Differences alone are not enough to reconstruct an entire video. The design also relies on original full reference frames. When a scene changes significantly, the encoder stores a full frame rather than only sparse events. These reference frames serve as Mainframes or eye frames, while the intermediate frames are reconstructed from synaptic differential events.

8. Storing Synaptic Data

    The project's target storage format is a codec structure called dotNVD or Neuromorphic Video Differential. Instead of storing conventional frame by frames of the video alone, the intent is to store mainframes, differential events, clustered motion data, loop references, and related reconstruction information in a compact dedicated, oriented windows file format. This is meant to make storage more efficient and also make the representation portable across compatible systems.

9. Homeo-gated control

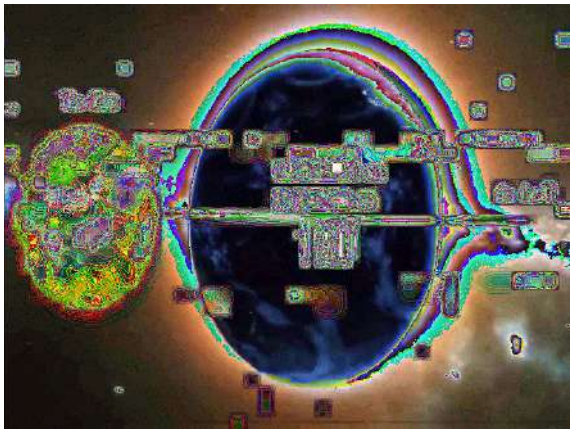
    The system is designed so that synaptic arithmetic is gated by actual spike activity rather than performed uniformly everywhere. In parallel chips like Cyclone IV, a supervisory control layer monitors global firing activity and adjusts bias values to prevent the network from becoming either inactive or unstable. I do have a broader interest in making the system a global standard in data transmission.



More than 70 times compressed, but the text is still readable!

Original Version (78MB) and dotNVD Compressed version(997KB)

I want to be equally clear about the project's present weaknesses. The compression logic is not final. Some reconstructions still show artifacts,



Artifacts of old pixel clusters are floating, causing massive frame instability

unstable output, inconsistent recognition quality, and fluctuating results across repeated trials. In some cases, text or fine structure is not reconstructed reliably. Some benchmark targets are still aspirational rather than fully met in hardware. The project includes many engineering risks, including bandwidth limits, memory constraints, clock-domain timing issues, and difficult tradeoffs between file size, quality, and stability. I do not consider those failures separate from the work; they are part of the work. They are the exact problems I am trying to understand and improve.

I would be grateful if Phillips Academy would read this update together with the user manual and, if available in my materials, the earlier theory document and linked demonstration media. The earlier white paper explains the larger conceptual direction; the user manual and this note explain the implemented system, the algorithms behind it, and the engineering challenges I am still working through. This submission represents the most complete version of the project I have produced so far.

Thank you very much for your time and for considering this additional material Please read the attached documents. This update was delayed late due to an unavoidable family circumstance.

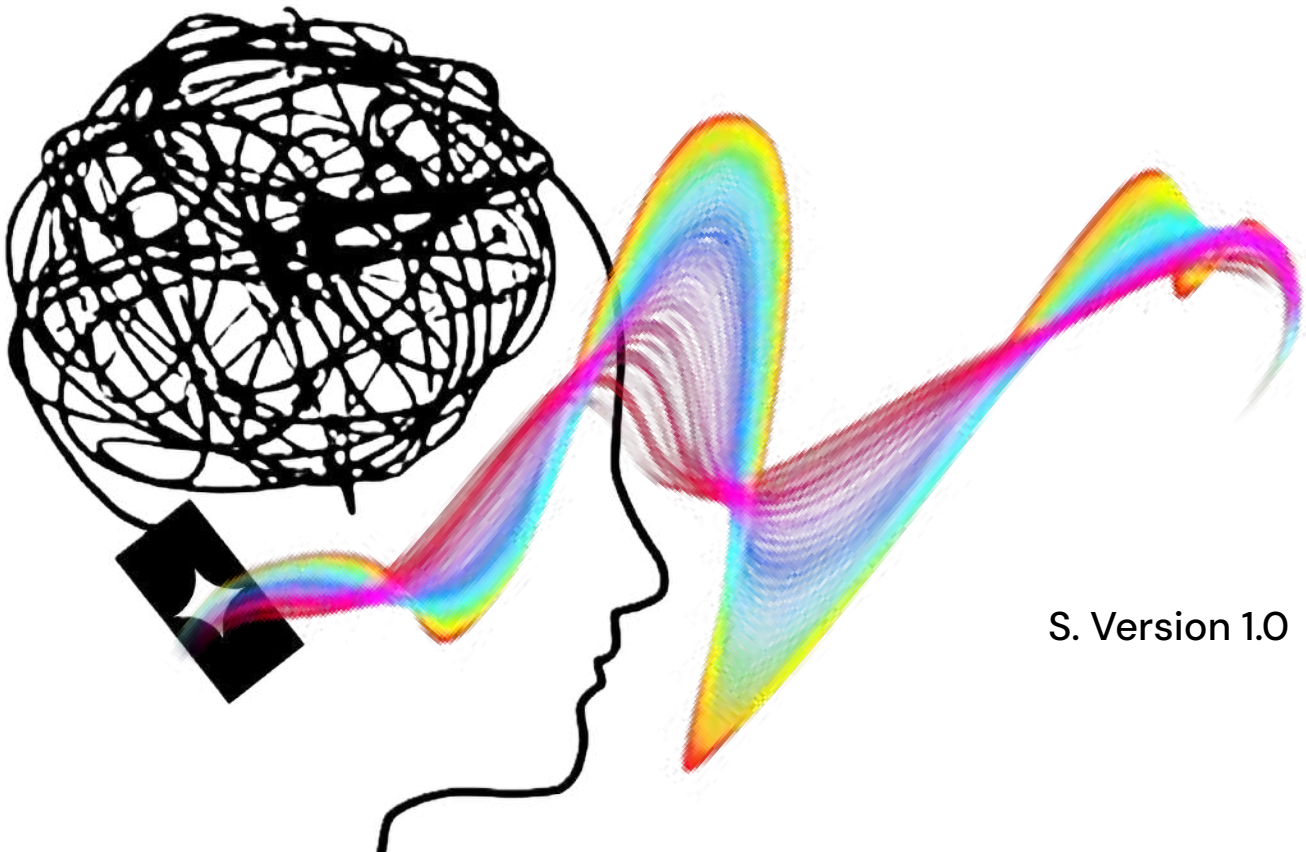
GVMA

NonPU™

The World's First Neuronal Processing Unit that is open and available to both consumers and industry.

User Manual & Core Docs for

✦Instant™



S. Version 1.0

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This document is issued for the 2026 Beta Release only and will be discontinued after March 10, 2026 (Anywhere on Earth Time).

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First Beta Release: January 2026
Open access.

This Product is Self-classified by GVMA Research Labs as safe for humans aged nine or above.

NonPU Instant is a hardware-assisted video compression and playback system developed by GVMA Research Labs. The system consists of three primary components: the NonPU Instant device, the NonPU Instant desktop application for Windows, and the .NVD file format. Together they form an integrated workflow designed to compress standard video into a compact, neuromorphic representation and to play that representation back through the dedicated hardware.

NonPU stands for Neuronal Processing Unit. The underlying concept draws from neuromorphic computing a field concerned with processing information in ways that reflect the event-driven, differential nature of biological neural systems. In practical terms for the end user, this means that NonPU Instant does not treat every video frame as equally important. Instead it identifies and encodes change, motion, and salient visual events, dramatically reducing file size while preserving the information needed for hardware-assisted playback.

This guide is intended to help users get the system running, understand the workflow from device setup to video export, and resolve common issues. Sections are arranged in the order a new user would typically encounter them.

The Neuromorphic Video Differential (dotNVD)

Files produced by NonPU Instant carry the .nvd extension, which stands for Neuromorphic Video Differential. The .NVD format is the native output of the NonPU Instant compression pipeline and serves as the primary medium for storing, sharing, and replaying content within the NonPU ecosystem.

How .NVD differs from conventional video

A conventional video file stores a sequence of full image frames, or a combination of reference frames and compressed differences between them, suitable for playback on any standard media player. An .nvd file is structured differently. Rather than reconstructing every frame independently, the format encodes differential information the changes between frames, along with saliency and habituation data that the NonPU hardware uses during reconstruction. This design keeps file size low and keeps the NonPU device central to the decoding process.

Because the format is built around hardware assisted playback, an .nvd file is not intended to open in a standard media player. The NonPU Instant application handles import, decryption, decoding, and sending the content back to the connected device for display.

The .NVD format is designed to reduce video file size dramatically, preserve the information the NonPU hardware requires for reconstruction, and keep the device an active participant in the viewing experience rather than a passive storage accessory. A shareable, encrypted variant of the file can be exported from the application and sent to other licensed users.

The official Windows® File Icon for dotNVD Files

What's in the Box

The NonPU Instant package includes all items needed to begin using the system. The contents described here reflect the standard configuration. Depending on version, upgrade package, or special request, the box may include additional materials.

Included items

NonPU Instant device

A multi-board hardware unit built around the NonPU processing system.
The device contains a main processor board and a secondary processing section.

Two USB Type C power cables

One cable is identified by an orange USB Type C connector head.
This cable is for the bottom main processor connection. The second white headed cable powers the slave processing section.

NonPU HDMI to VGA converter

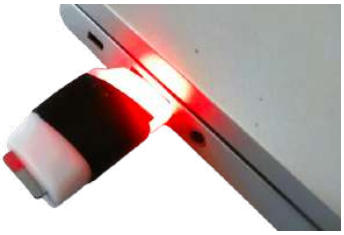
For displays that accept VGA input rather than HDMI.

NonPU Instant Drive

A 64 GB microSD based storage medium.
The drive holds license data and is prepared using the NonPU Instant application.

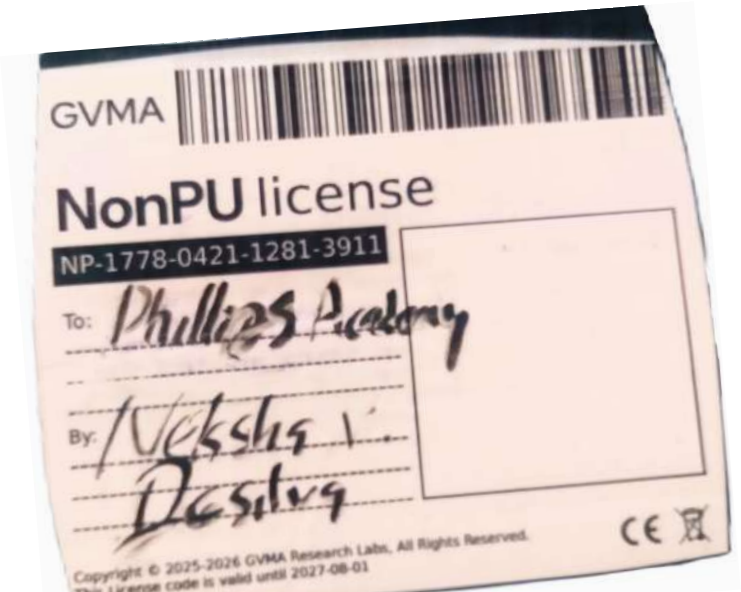
USB Type A to microSD reader

Included as a tool for Computers without built in MicroSD card Port,



User manual

This guide, or printed equivalent, covering setup, operation, and troubleshooting and license information



4800 GigaBytes? But how?

The NonPU Instant device ships with a 64 GB SD card. In raw terms, that is 64 gigabytes of physical storage. However, the effective storage capacity measured in terms of the original video data it can represent is approximately 4,800 gigabytes.

This estimate comes from the compression benchmark observed during testing. A 720p, 10 minute MP4 video of approximately 98 megabytes was compressed by the NonPU algorithm to about 1.98 megabytes. This corresponds to a compression ratio of roughly forty nine to fifty times smaller than the original file. This test case was selected because it reflects a common type of video found on the internet. Many online videos fall within the 10 to 15 minute range and are commonly encoded at 720p resolution. At typical bitrates, a 10 minute 720p MP4 file usually occupies between 80 and 100 megabytes, which makes the 98 MB sample representative of real world content rather than a specially selected best case.

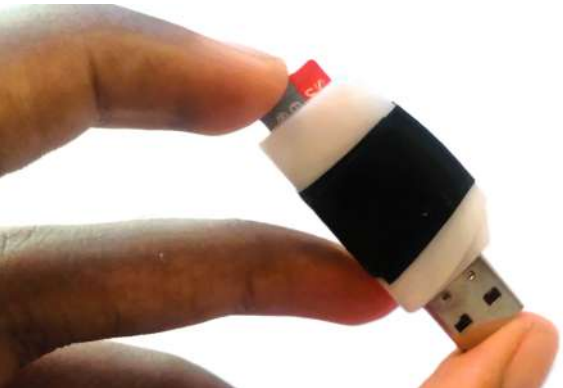
When the measured compression ratio is applied to the full 64 GB storage card, the represented amount of original video data increases significantly. At the observed ratio of roughly fifty to one, the sixty four gigabyte storage capacity corresponds to approximately three thousand one hundred sixty five gigabytes of original video.

Across a wider set of content, the broader median compression ratio measured during testing was approximately fifty two to one. At that ratio, the sixty four gigabyte storage capacity represents about three thousand three hundred twenty eight gigabytes of original video data.

In some cases the compression ratio reached values near sixty two to one. Under those conditions the same sixty four gigabyte storage capacity corresponds to approximately three thousand nine hundred sixty eight gigabytes of original video data.

The estimate of approximately 4,800 gigabytes represents a high end projection. This projection takes into account that MP4 files are already compressed formats. When the algorithm operates on video data before conventional MP4 encoding is applied, the effective compression relative to the underlying source material becomes higher, which increases the equivalent storage representation.

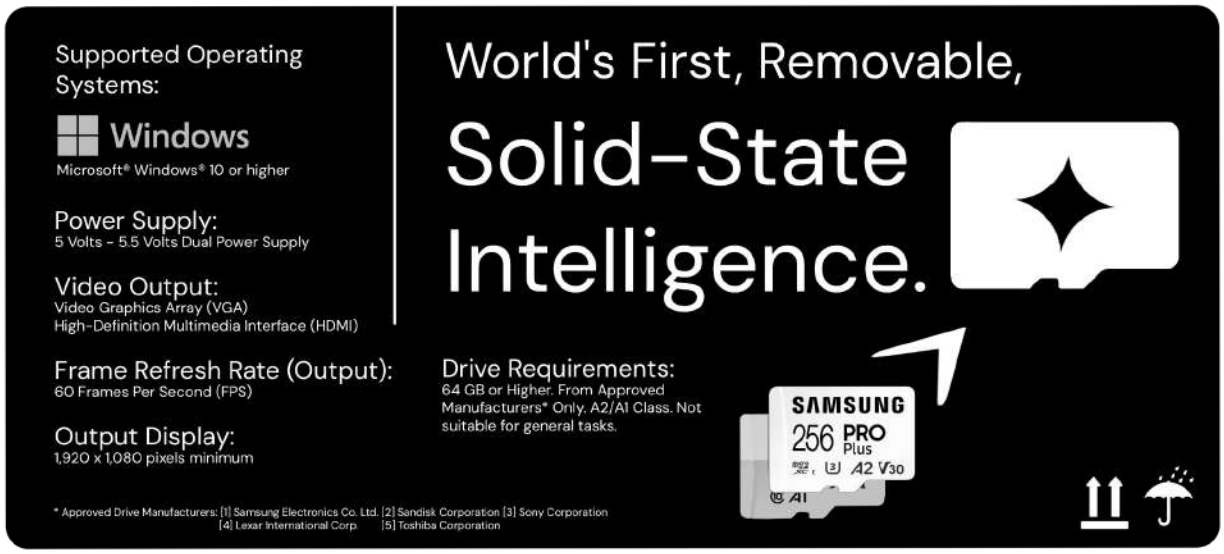
The practical summary is that a 64 GB NonPU Instant drive can represent roughly 3,300 to 4,800 gigabytes of original video depending on the characteristics of the input footage, with measured compression ratios typically ranging between about forty nine and fifty seven times smaller than the original files.



Before You Begin

Before using NonPU Instant for the first time, certain conditions must be met. Having each of these in place before powering the device avoids interruptions during the initial setup flow.

Requirements



Refer to the back of the box

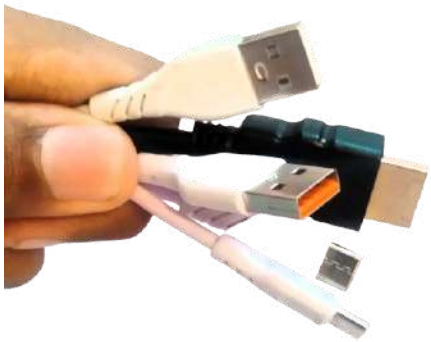
A display with HDMI input is required, connected directly to the NonPU Instant device. If the display accepts only VGA, the included NonPU HDMI to VGA converter bridges the connection. A USB mouse is needed for on-device interaction through the connected display.

A Windows computer running Windows 10 or later, with at least 8 GB of RAM, available USB connectivity, and administrator access where possible, is required for the NonPU Instant desktop application. The setup flow, device connection dialog, and drive preparation wizard are each designed specifically for Windows.

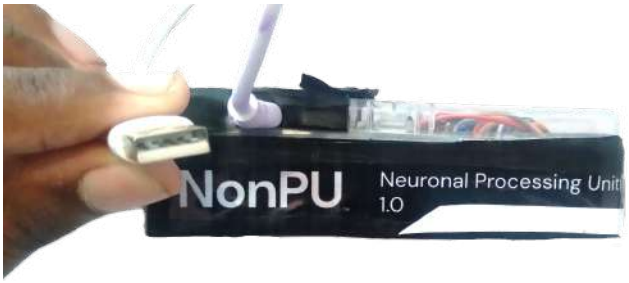
The NonPU Instant Drive, the included microSD card must be available. It is used during license verification and should not be in use by another device when the setup flow begins.

Both USB Type C power cables must be ready before the device is powered for the first time. Powering only one port is not recommended unless GVMA Research Labs specifically instructs otherwise.

Powering the Device



NonPU Instant requires both USB Type C power connections to be active simultaneously. The device follows a dual-board architecture in which the main processor board handles primary memory and control, while the secondary board assists with parallel computation. Each board has its own power input.



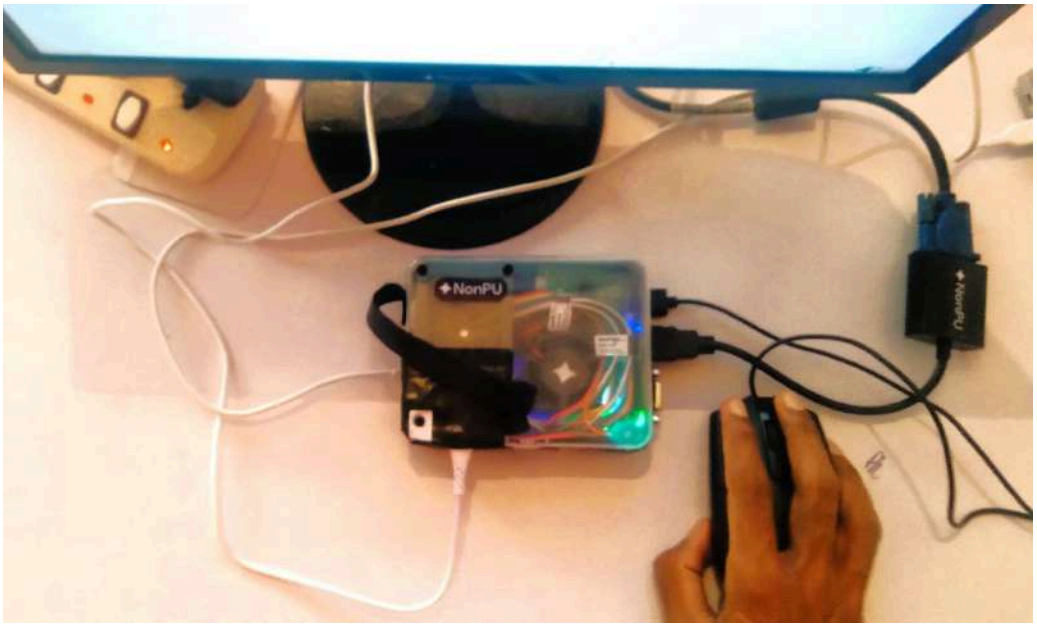
The orange headed USB Type C cable connects to the bottom port, which serves the main core processor. The second cable with a standard connector head connects to the upper port for the secondary section. Both cables should be connected to their respective power sources before the system is considered powered on.

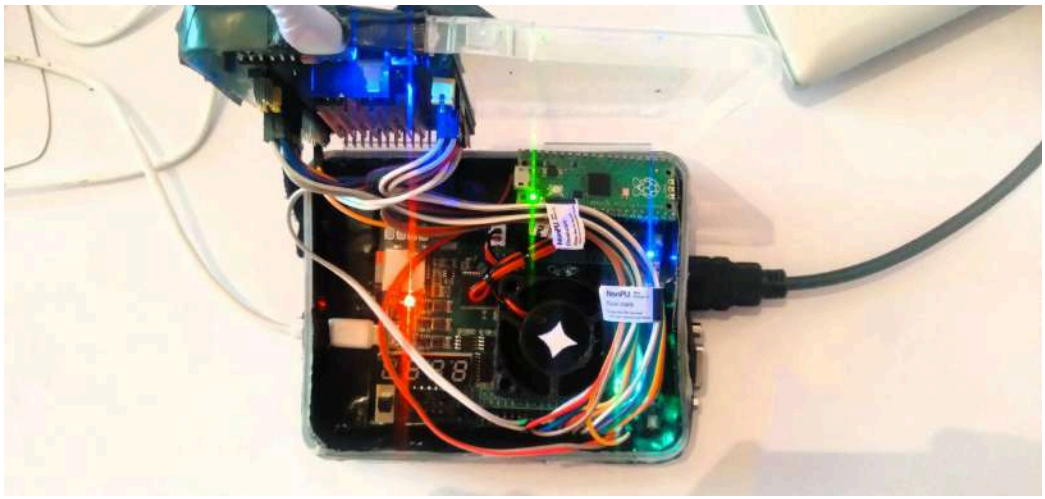
Do not power only one side of the device during normal operation. Do not disconnect either power cable while the system is actively running, transferring, or preparing data.

Connecting a Display and Mouse

Once the device is powered, connect a monitor to the NonPU Instant device using an HDMI cable. If the monitor requires a VGA signal, use the included NonPU HDMI to VGA converter between the device and the monitor.

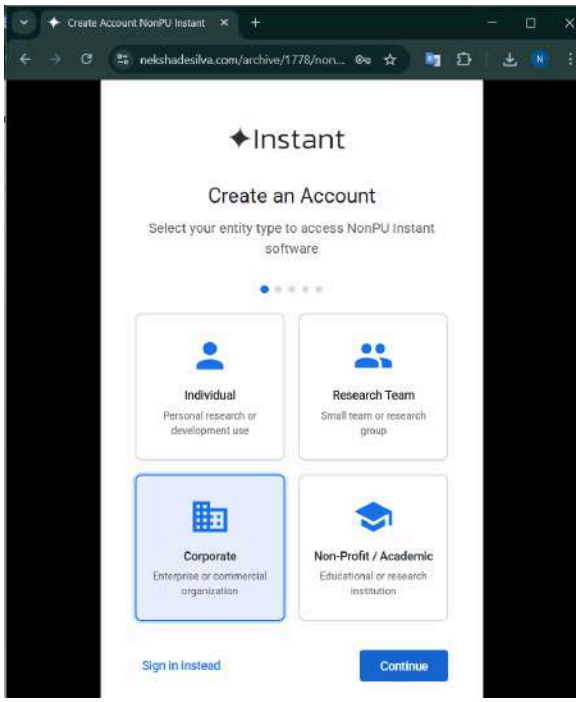
Connect a USB mouse to the USB port adjacent to the display output. Turn on the display. The device provides a visual interface through the connected display that allows the user to interact with playback controls and other on-device functions using the mouse.

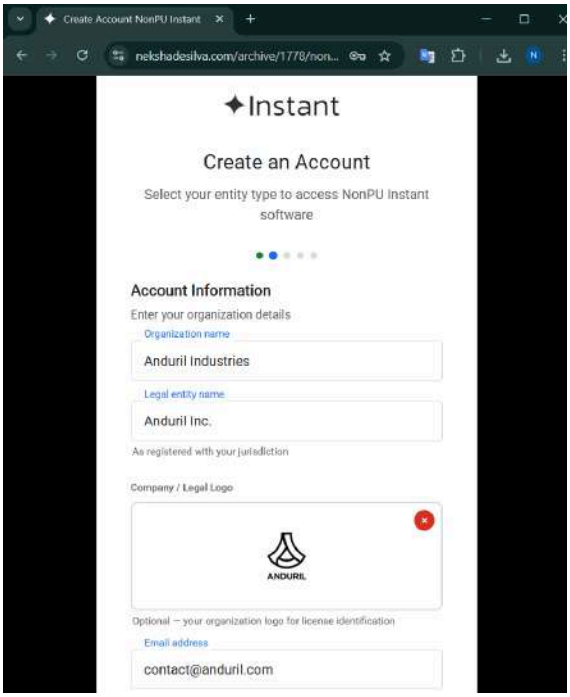




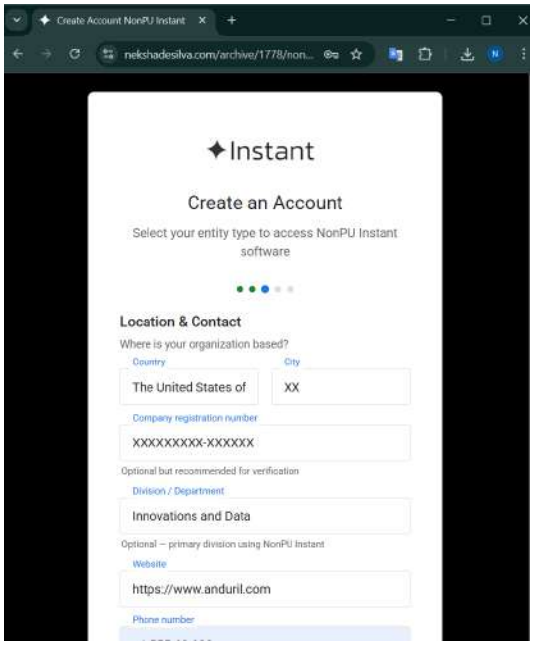
Creating an account

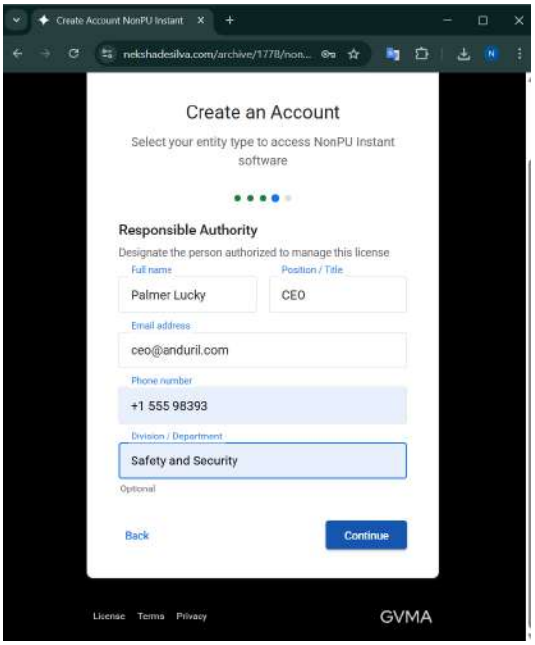
Visit www.nekshadesilva.com/archive/1778/non-pu,
Choose your entity type and fill in the initial information to continue.

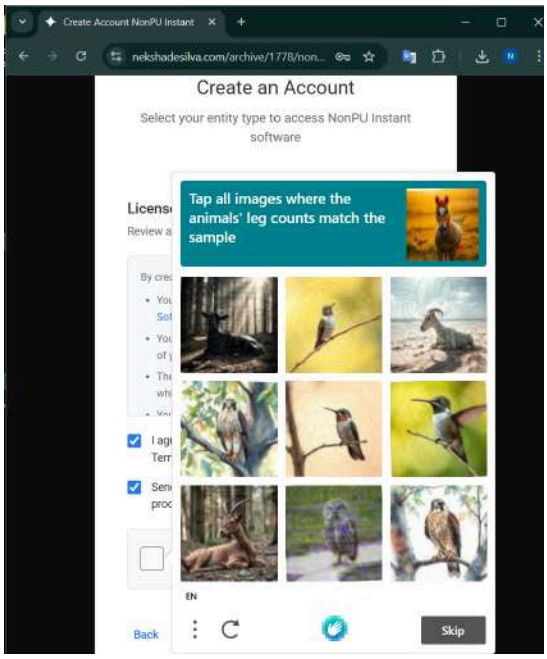


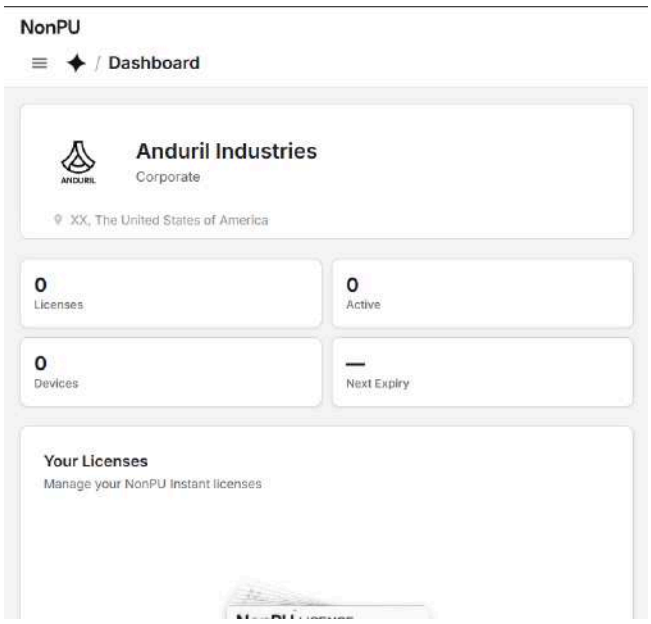


Verify the reCAPTCHA, provide your personal or corporate details, Upload a Corporate/Academic/ Personal Profile Image, then complete the registration.









After completing your account creation, your web dashboard will look like the interface shown above.

After the NonPU Instant device has been successfully flashed with a valid license key, the device will emit a blue indicator light. The blue indicator remains active until the license expires according to the internal clock, or until the storage drive is removed or replaced.

⚠️ This dashboard is connected to the system but operates separately from the NonPU Instant Desktop Application dashboard. Avoid any confusion.

Installing NonPU Instant for Windows®

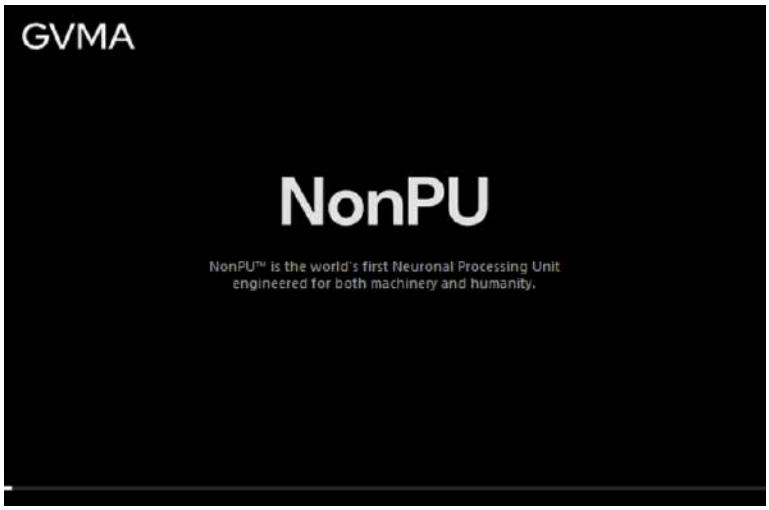
The NonPU Instant desktop application is installed on a Windows computer and handles video compression, drive preparation, license management, export, and device communication. The application is available at

www.nekshadesilva.com/archive/1778/non-pu/downloads

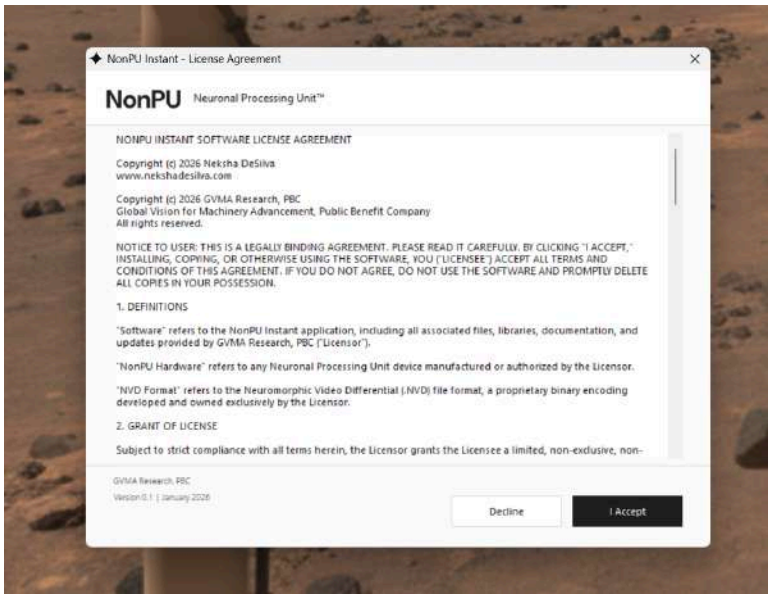
After downloading the installer, run it and follow the on screen steps.

Splash Screen

Please wait while the software checks for Win32 system errors and analyzes your computer specifications.



Terms and Privacy Policy agreement



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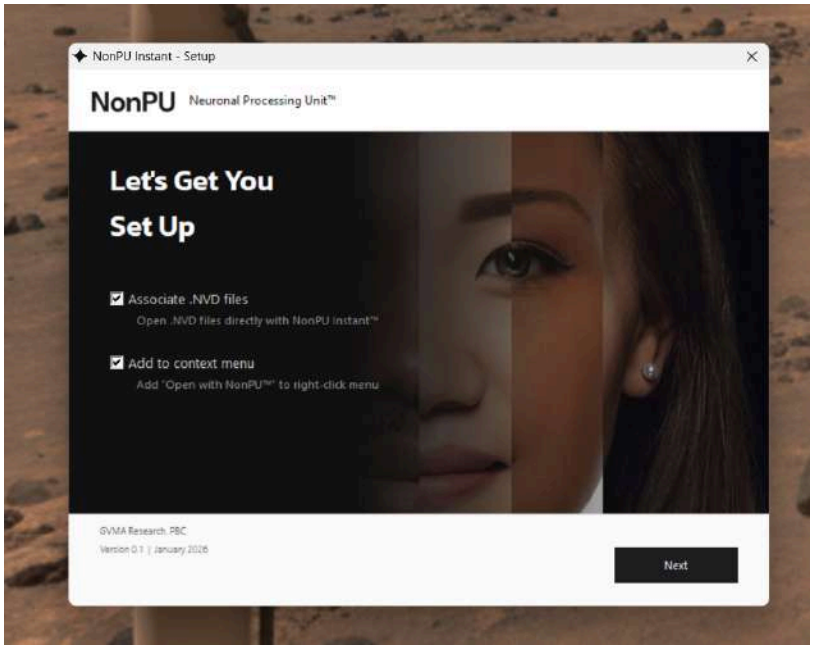
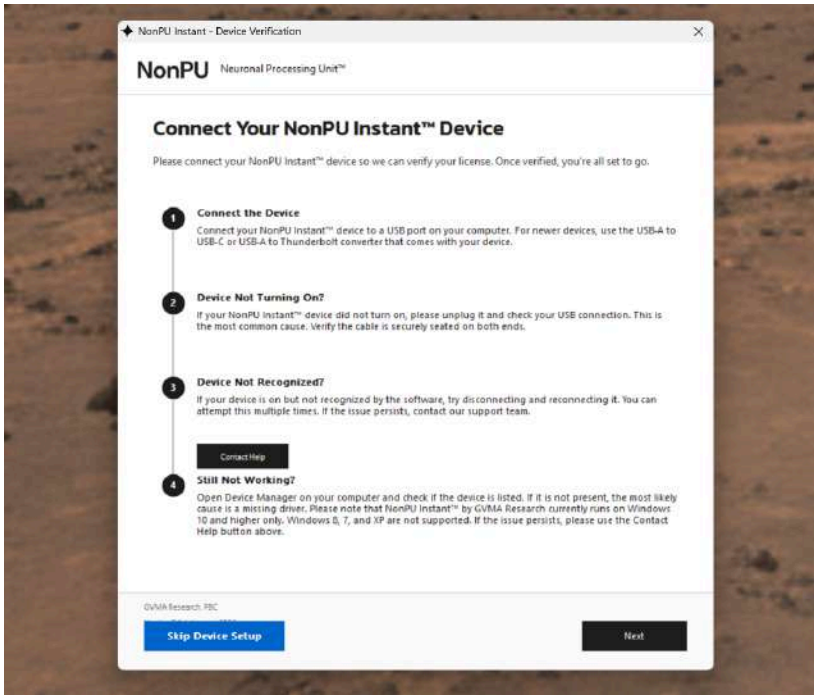
dotNVD file permissions. Please check these permissions to create NVD files, edit and register them in the Windows Registry.



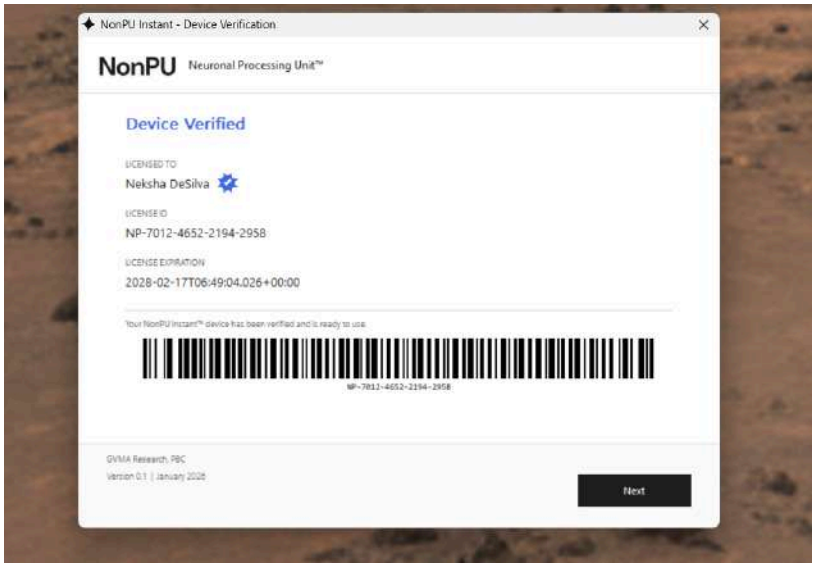
Connect the NonPU Instant Device

To continue with the configuration, connect your NonPU Instant Device. The Windows software will validate the license stored on the device. If the license is missing or expired, follow the instructions in the [License Registration or Renewal](#) section.



Validation

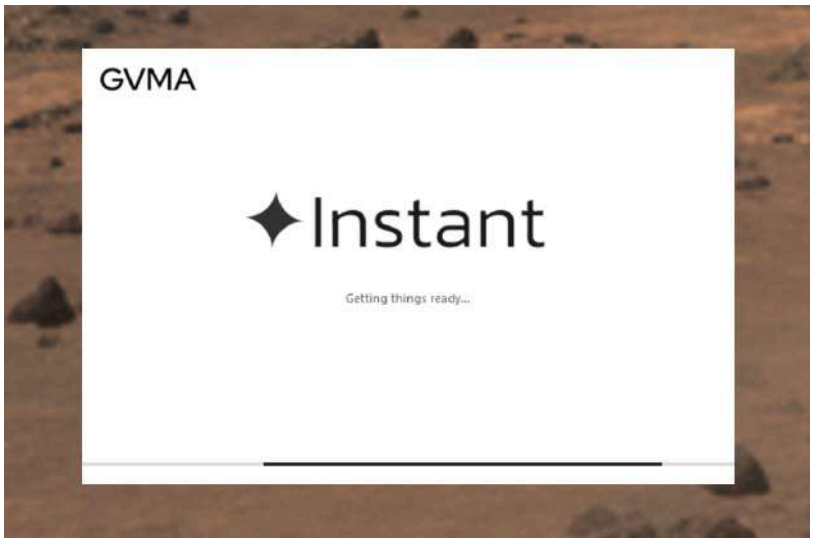
After the device is validated, this interface will display your license key, signed in account details, license information, barcode, and the license validity date.



Driver Installation & Internal Application Processing

For transparency, the software installs several required drivers on your computer (12 in total), including:
Microsoft DirectX 12, CH350 Serial Driver, Intel USB Blaster Driver, and Windows Foundation Driver.

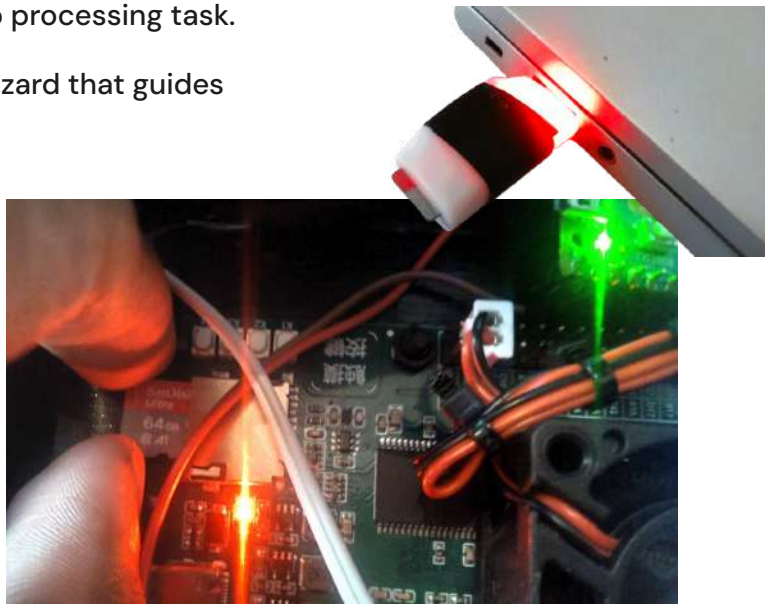
To view the complete list of drivers, please visit
nekshadesilva.com/archive/1778/non-pu/downloads



Licensing the NonPU Instant Device (Self service)

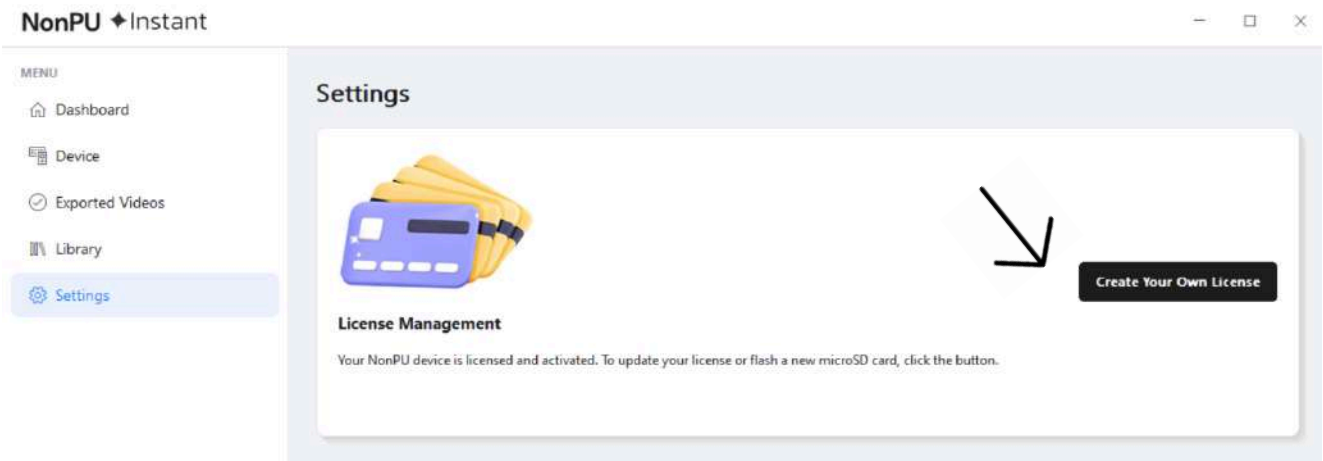
The NonPU Instant Drive is the microSD card included in the package. It holds license data, synaptic data, dotNVD files, Partitioning data, metadata and must be prepared using the NonPU Instant application before the first video processing task.

The application includes a dedicated license wizard that guides the user through this process.

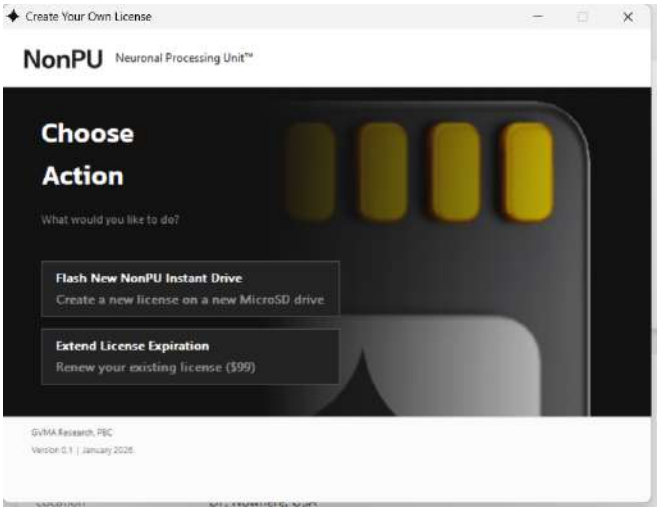


Preparing the drive for the first time

Remove the NonPU Instant Drive from the device if it is currently inserted.
Insert the drive into the included MicroSD to USB type A converter or a compatible card reader, and connect the adapter to the Windows computer.
Open NonPU Instant and follow the license or drive preparation wizard.



Depending on the current state of the drive, the wizard may flash a new NonPU Instant Drive, create a new license, extend an existing one, or prepare, sign, and format the drive before proceeding. The wizard may restart the application after the process completes.

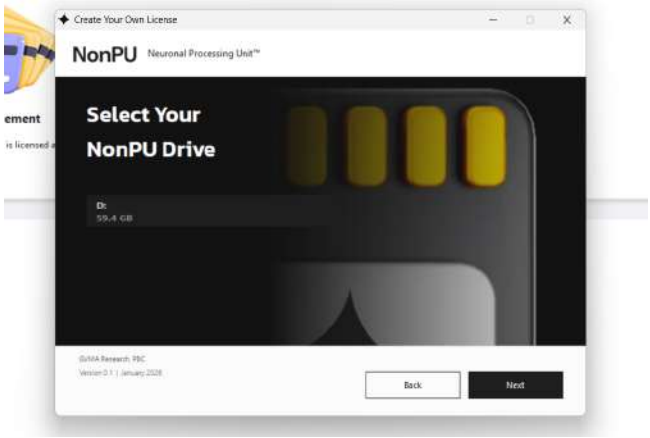


Select Your NonPU Instant Drive

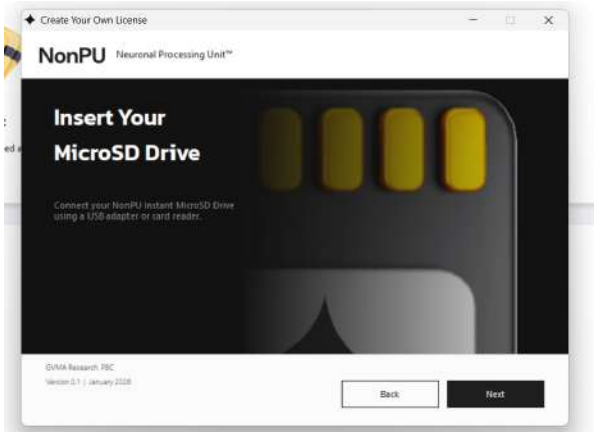
In this step, you need to select your microSD card or NonPU Instant Drive from the list provided.

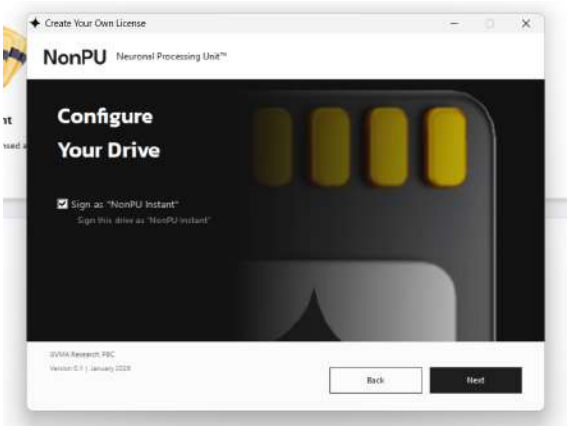
Please note that this list may also display other connected storage devices, such as USB flash drives, external drives, or connected mobile phones. Because of this, it is important to carefully verify the correct drive before proceeding.

⚠️ The selected drive will be fully formatted, which will erase all existing data on that drive.

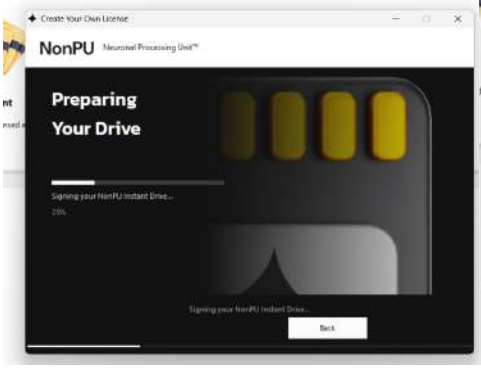


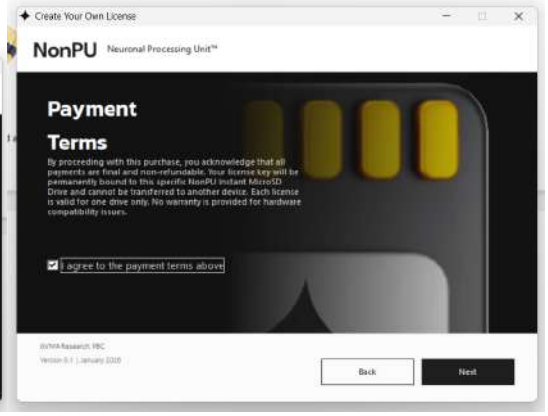
Move Ahead!

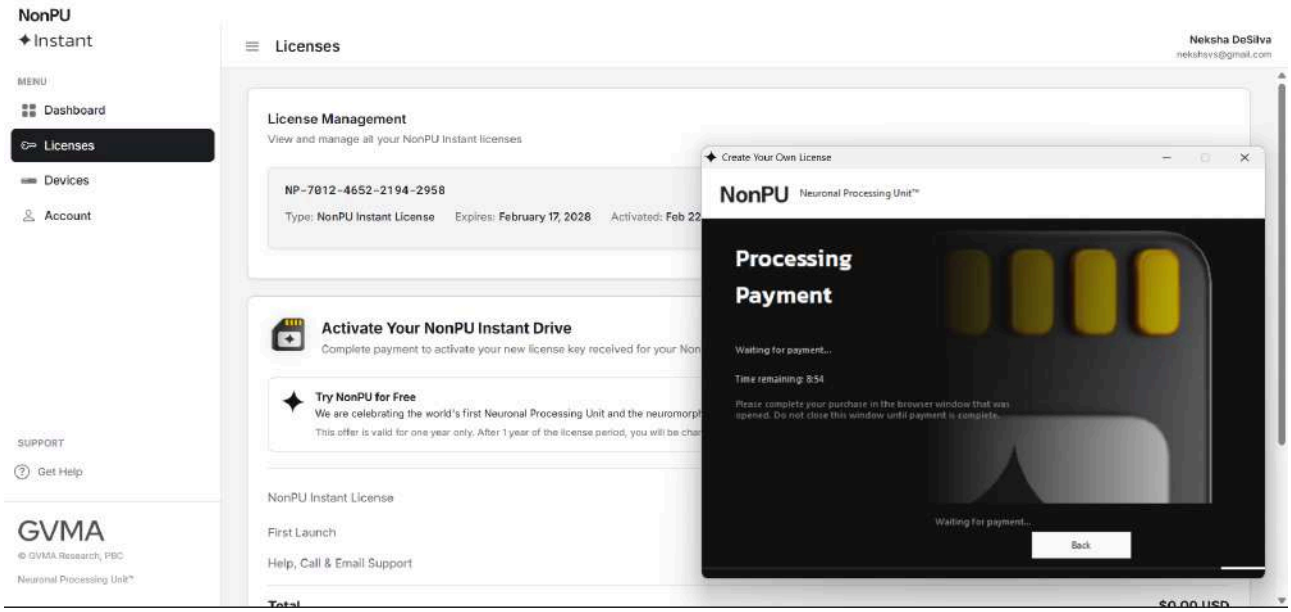


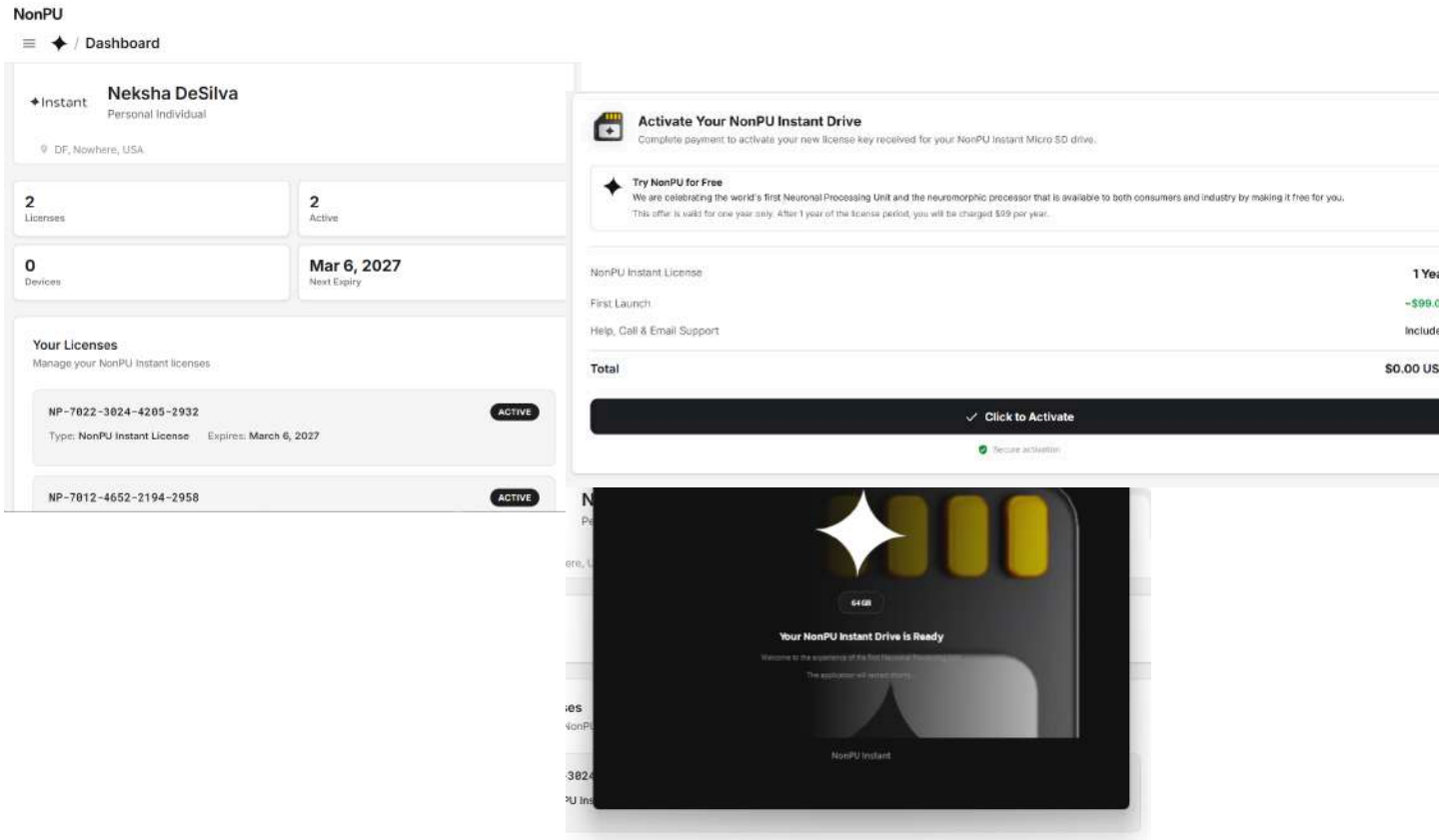


In this step, you will be redirected to the payment section. Your NonPU Instant Drive will be formatted during this process.









⚠️ **Important** Use the included adapter when preparing the drive. Do not remove the NonPU Instant Drive while the application is actively preparing, flashing, signing, or validating it. The wizard includes checks for premature drive removal and will terminate the procedure if the drive is disconnected before the relevant stage is complete. If this occurs, reconnect the drive and restart the process from the beginning.

Compressing a Video for the First Time

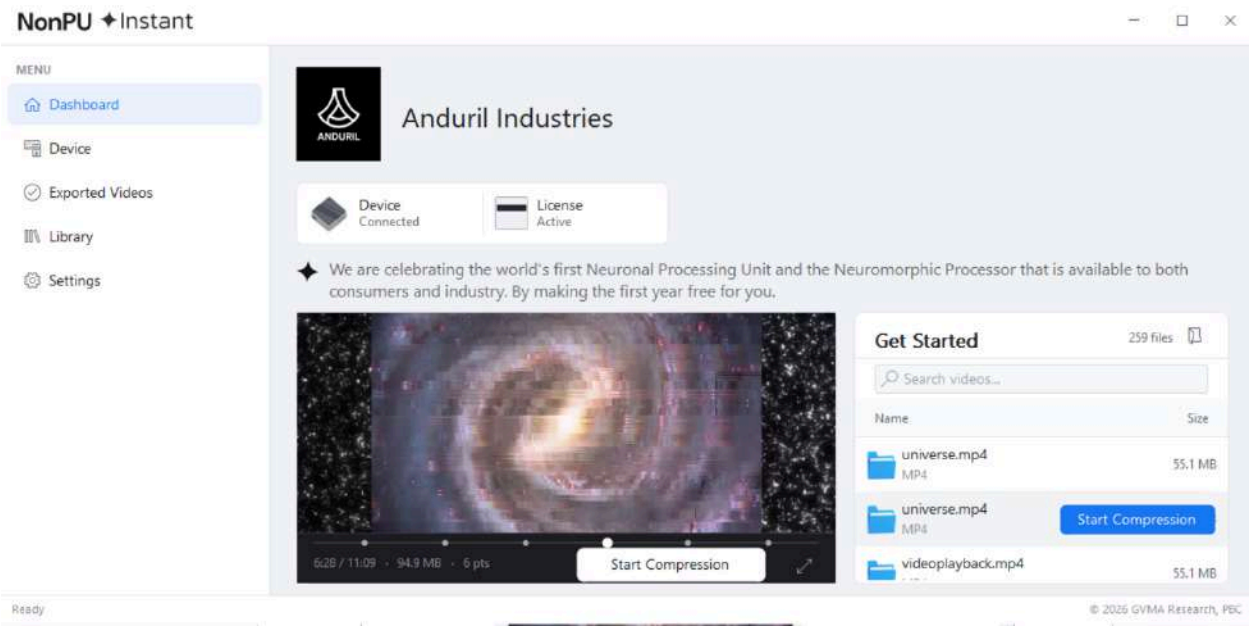
With the device connected, the license verified, and the application installed, the user is ready to compress a video and produce an .nvd file. The process requires any of your busy beaver videos, a connected NonPU Instant device, and a valid license.

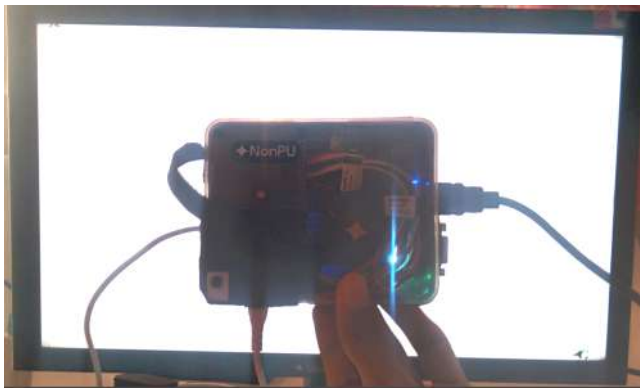
⚠️ If any of these conditions are not met, the application will indicate which step needs attention before compression can begin.

Loading a source video

From the main application dashboard, use the file selection control to browse to the source video. The application accepts common video formats including .mp4, .avi, .mkv, .mov, and .wmv. Once a file is selected, the application automatically generates an output path using the .nvd extension. The output path can be adjusted if needed.

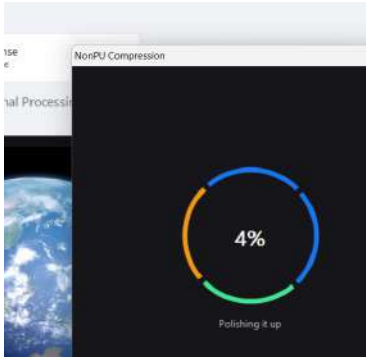
Now, let us select a video of 95 Megabytes to compress using our Algorithmic Logic.





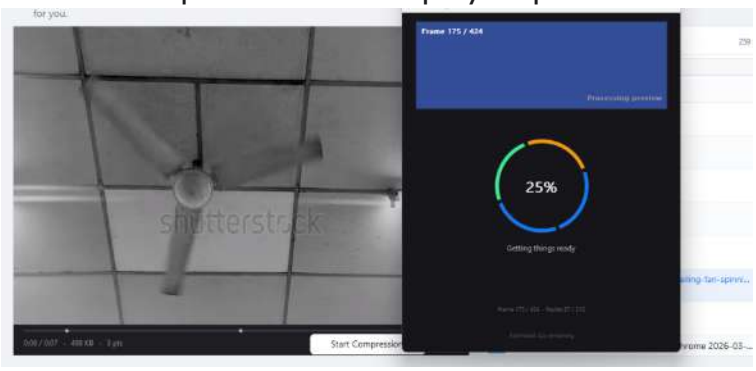
Compression Stage 1,2 and 3

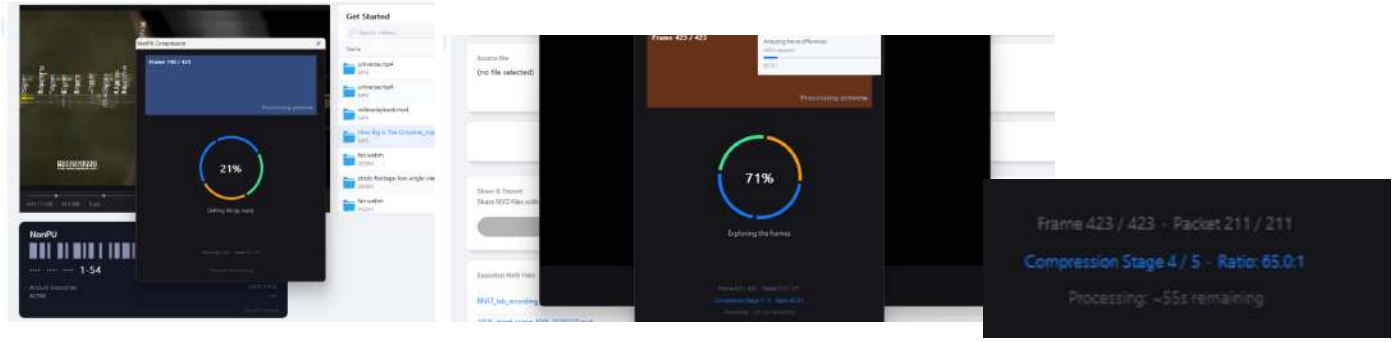
With a source file loaded and all conditions met, clicking Start Compression begins the NonPU Instant compression workflow. The application moves through a multi-stage pipeline, tracking upload progress, per-stage compression ratio changes, live telemetry, and estimated time remaining. In the user interface, this process can be understood as uploading the video to the NonPU pipeline, analyzing differential changes between frames, refining salient information, and assembling the final .NVD output.



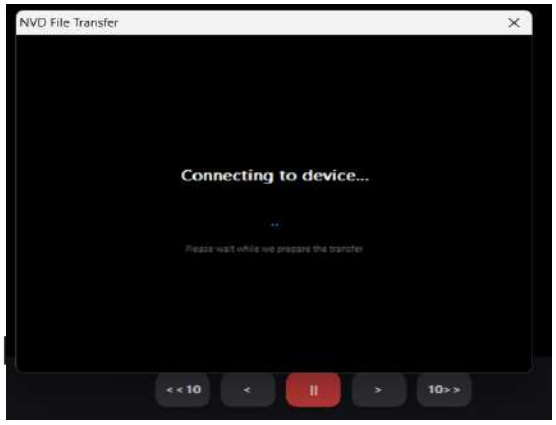
Compression Stage 4 and 5

Stage 4 begins after the desktop application has sent the video data to the NonPU Instant device. The device stores the incoming data as temporary track files. The number of track files is not fixed because it depends on the file format, file size, and the amount of data being processed. These track files are loaded into SDRAM in manageable groups so the device can process them step by step.



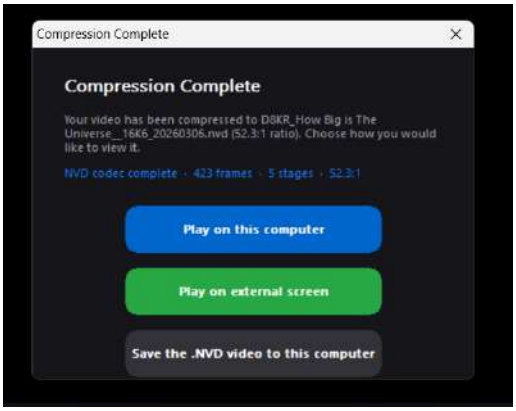


At this stage, the system prepares the video for NonPU Instant compression. Since the original MP4 file is already compressed, the device first expands the video data into an intermediate RAW565 format so the full RGB data can be processed. This intermediate data can become much larger than the original video file. Because of this, the SD card is used for one active encoding process at a time. The device then processes the data group by group and converts it into the final .NVD file using its onboard processors.



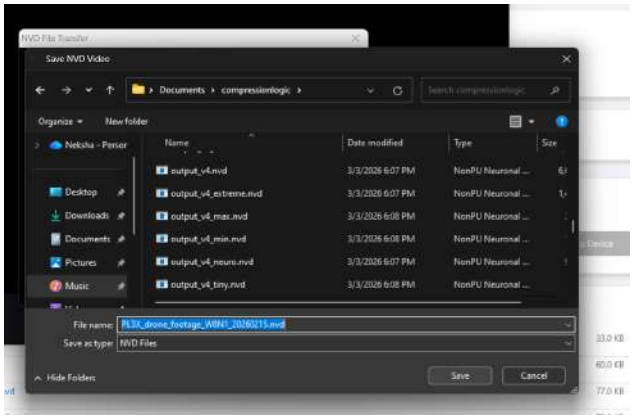
Stage 5 is running parallel with Stage 4. While the device is processing the video, the desktop application shows live progress updates for each stage. This allows you to monitor the status in real time.

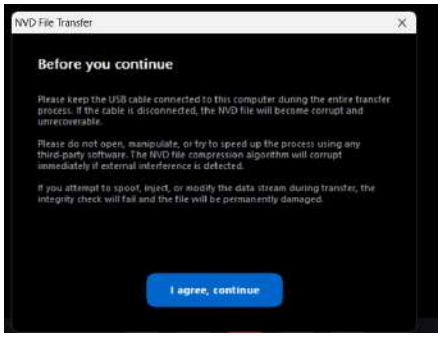
When processing is complete, the dotNVD file is available on the NonPU Instant device. You can then choose how to use it.

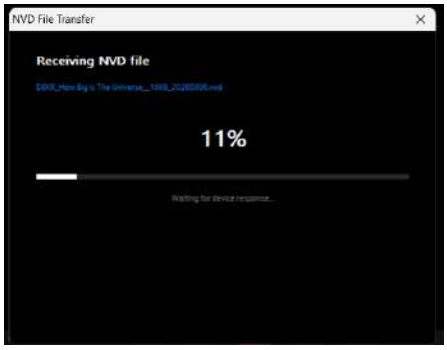


You can play it through a connected monitor or if an external dedicated monitor is not available, Play it on your own device, depending on your setup. Because the .NVD file is smaller than the original MP4, transferring it is faster than the original upload.

You can Save the dotNVD file to a known local destination or play it directly using an External Monitor.

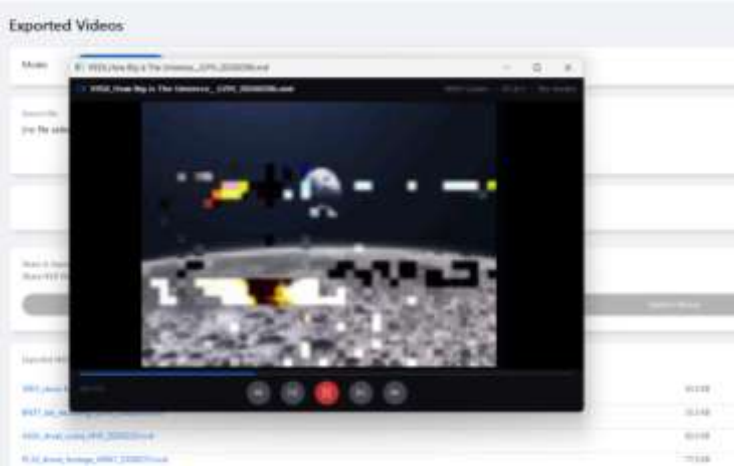






Playing the dotNVD video on an External Device (1.78 KB)

Playing the dotNVD video on-Device



Changing the Compression Quality Manually

This setting allows you to adjust the compression quality and sample rate parameters used during the encoding of .nvd video files.

One of the main parameters in this section is the Keyframe Interval, which is closely related to the frame rate (FPS) in the NonPU Neuronal Video Differential technology. The keyframe interval determines how frequently keyframes appear in the video stream and therefore strongly influences the effective frames per second.

If you want to increase the FPS, you can increase the keyframe interval using the plus icon. If you want to reduce the FPS, you can decrease the keyframe interval using the minus icon.

The Compression Quality control determines the balance between video quality and file size of the generated .nvd file. Increasing compression intensity will reduce the storage size of the .nvd file but may slightly reduce visual quality. Reducing compression intensity will increase the video quality, but the resulting file will require more storage space.

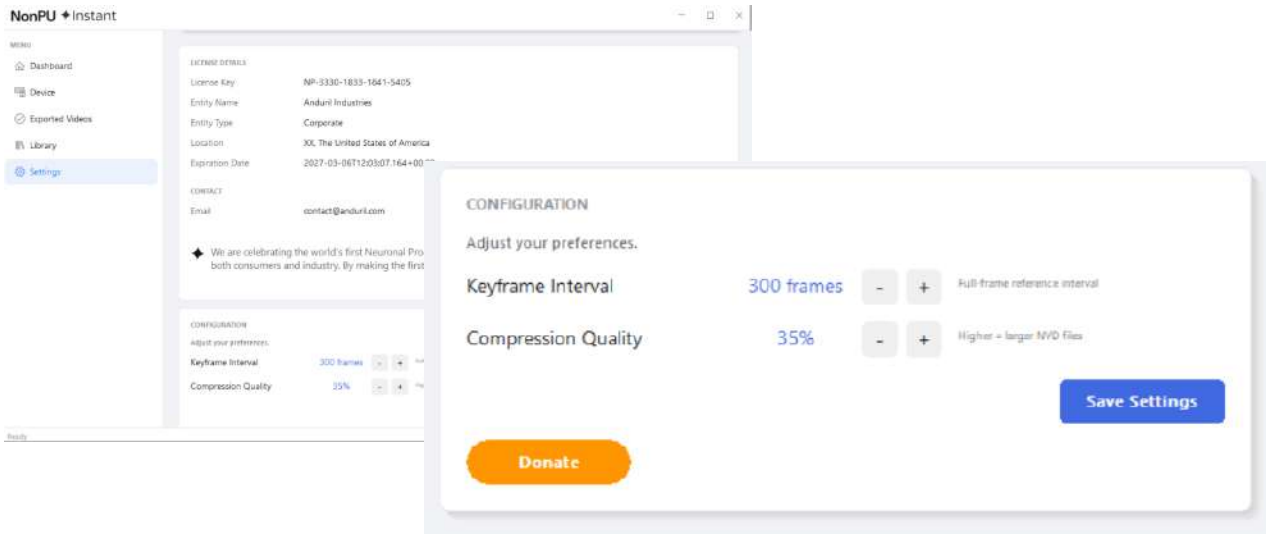
Please note that modifying these settings can be considered a destructive configuration change, as the selected values remain active for future encoding operations until they are manually changed again. These parameters directly affect the encoding and decoding process of the .nvd file format, and incorrect adjustments may lead to reduced performance or unexpected results.

By default, the recommended values are

Keyframe Interval to 300 frames

Compression Quality to 35%

These default values are optimized for balanced performance, quality, and file size. If you are unfamiliar with these parameters, it is recommended to keep the default settings.



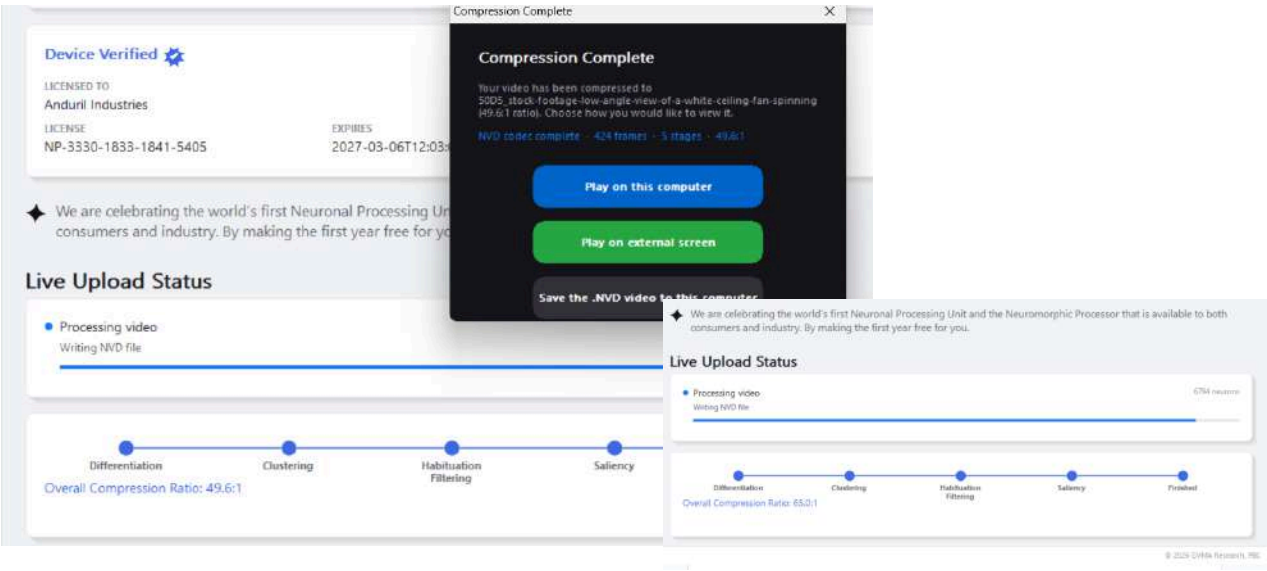
Device Status Indicators

The NonPU Instant device indicator area will turn full green while compression is in progress, indicating that the device is operating normally. ilf an issue occurs during operation, the indicator area will display different light patterns to signal the type of error.

1. A red and green blinking light indicates that the media file being processed is corrupted or cannot be decoded.
2. A blinking red light indicates a device hardware failure or a master device SDRAM overload.
3. If the green light blinks while the red light remains solid, it indicates an incompatible software version or unrecognized signals. This situation may also occur if the USB ports are damaged or dusty, the device firmware is outdated, drivers were installed manually, or the operating system is incompatible.

Completion

When the process finishes, the application reports the path of the encoded output file and the compression ratio achieved. If licensing is not yet complete when compression is attempted, the interface redirects the user to the license purchase settings page rather than beginning compression.

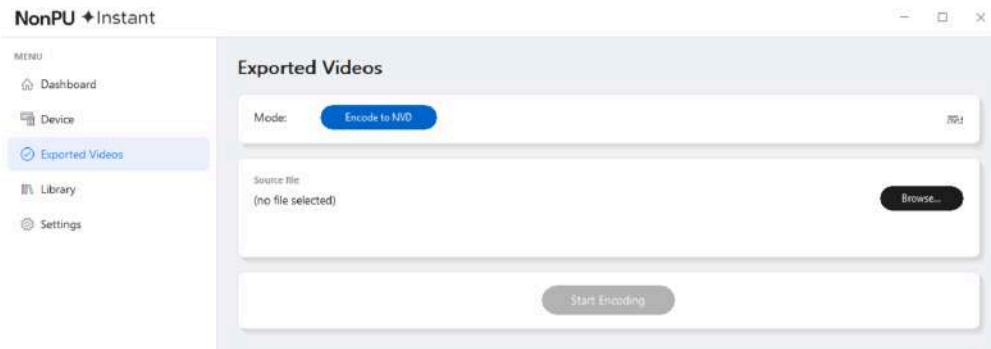


Exporting and Locating .NVD Files

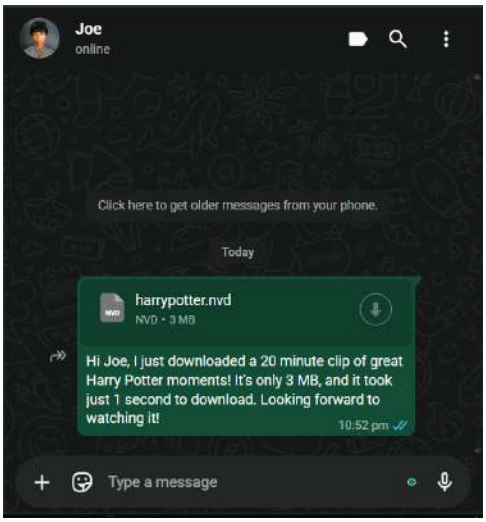
After compression completes, the .nvd file is saved to the output location selected during setup, or alongside the source file if the automatically generated path was kept. The application displays the final output path on the completion screen.

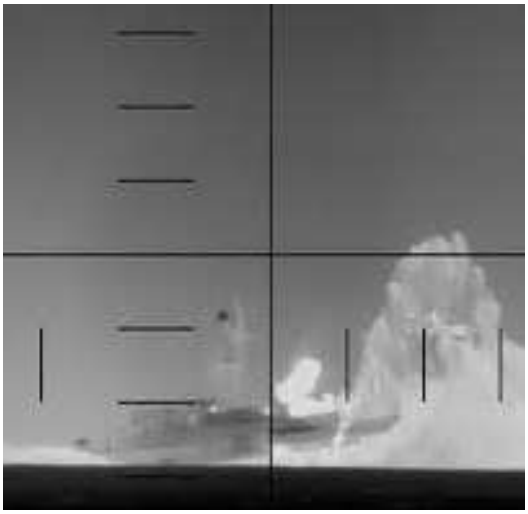
Exporting for sharing

The application provides an Export Encrypted function, labeled Export Encrypted NVD for Sharing, which saves the result as a shareable, encrypted .nvd file. This file can be transferred to another licensed user who can then import it through the NonPU Instant application on their own system.



#dotNVDeverywhere





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